End-to-End Testing of Airbnb Clone Web Application

Test Plan

1.1 Objective

To validate the functionality, usability, API behaviour, and basic performance of the Airbnb Clone web application, and to confirm that core user journeys work reliably for registration, login, property discovery, booking, payment, and profile management.

1.2 Scope

The testing scope includes the following modules:

- User Registration and Login
- Property Search and Filters
- Property Details Page
- Booking System
- Payment Flow
- User Profile Management
- **1.3 Out of Scope**
- **Third-party payment gateway settlement verification**
- Admin panel workflows
- Deep security testing
- **1.4 Testing Types**
- Functional Testing
- UI Testing
- API Testing
- Regression Testing
- Usability Testing
- Basic Performance Testing
- Compatibility Testing

1.5 Tools Used

- Manual Testing
- Postman for API testing
- Selenium with Python for UI automation
- **1.6 Test Approach**

. Review the application modules and identify critical user flows.

. Execute manual test cases for positive, negative, and edge scenarios.

. Record defects with severity, priority, reproduction steps, and expected versus actual results.

. Automate two critical scenarios using Selenium.

. Validate two API endpoints in Postman using status code and response body checks.

. Re-test impacted modules after fixes as part of regression testing.

Test Cases

TC ID	Module	Test Case	Preconditions	Steps	Expected Result	Type
TC-01	Registration	Verify user can register with valid data	User is on registration page	Enter valid name, email, phone, and password, then click Register	Account is created and success message is displayed	Positive
TC-02	Registration	Verify registration with invalid email format	User is on registration page	Enter invalid email such as usermail.com and submit	Validation error is shown for invalid email	Negative
TC-03	Registration	Verify registration with mandatory fields left blank	User is on registration page	Keep required fields empty and submit	Required field validation messages are displayed	Negative

TC ID	Module	Test Case	Preconditions	Steps	Expected Result	Type
TC-04	Login	Verify login with valid credentials	Registered user exists	Enter valid email and password, then click Login	User is logged in and redirected to dashboard or home page	Positive
TC-05	Login	Verify login with incorrect password	Registered user exists	Enter valid email and wrong password, then click Login	Error message is displayed and login is blocked	Negative
TC-06	Login	Verify login with unregistered email	User account does not exist	Enter unregistered email and password, then click Login	Proper error message is displayed	Negative

TC-07	Search	Verify property search with valid location	Properties exist for searched location	Enter a valid city or location and click Search	Relevant property listings are displayed	Positive
TC-08	Search	Verify search with empty input	User is on search page	Leave search field empty and submit	Validation message appears or default search behavior is handled gracefully	Negative
TC-09	Search	Verify search with no matching results	No property exists for search input	Search for unsupported location	No-result message is displayed without page crash	Edge
TC-10	Filters	Verify price filter functionality	Search results are available	Apply min and max price filters	Results refresh to match selected price range	Positive
TC-11	Filters	Verify date filter functionality	Search results are available	Select check-in and check-out dates and apply filter	Results show properties available for selected dates	Positive

TC ID	Module	Test Case	Preconditions	Steps	Expected Result	Type
TC-12	Property Details	Verify property details page displays correct information	Property listing exists	Open any property from search results	Title, images, location, price, amenities, and availability are shown correctly	Positive
TC-13	Booking	Verify booking with valid data	Logged-in user and available property exist	Select valid dates, guest count, and click Reserve	Booking is created successfully	Positive
TC-14	Booking	Verify booking attempt without login	User is not logged in	Select property and attempt to book	User is redirected to login or sign-up page	Edge

TC-15	Booking	Verify booking with invalid date range	Logged-in user and property exist	Select check-out earlier than check-in and continue	Validation error is displayed	Negative
TC-16	Booking	Verify duplicate booking prevention	Existing booking already blocks same dates	Attempt to book same property for overlapping dates	Booking is prevented and conflict message is displayed	Edge
TC-17	Payment	Verify successful payment flow	Booking details are valid	Proceed to payment with valid payment data	Payment succeeds and booking confirmation is shown	Positive
TC-18	Payment	Verify failed payment flow	Booking details are valid	Enter invalid or declined payment details	Payment failure message is shown and booking is not confirmed	Negative
TC-19	Profile	Verify user profile update	Logged-in user exists	Update editable profile details and save changes	Updated details are stored and visible after refresh	Positive
TC ID	**Module**	**Test Case**	**Preconditions**	**Steps**	**Expected Result**	**Type**
TC-20	Session	Verify logout functionality	Logged-in user exists	Click Logout button	Session ends and user is redirected to login or home page	Positive
TC-21	Session	Verify session timeout handling	Logged-in user exists	Stay inactive until session expires	User is logged out safely and prompted to sign in again	Edge
TC-22	Performance	Verify major pages load within acceptable time	Stable internet connection	Open home, search, property, and booking pages	Each major page loads within 3 seconds under normal conditions	Performance

TC-23	UI	Verify mobile responsiveness	Application is opened in mobile viewport	Open key pages in responsive mode	Layout adjusts properly with no overlapping or hidden elements	UI
TC-24	UI	Verify navigation menu links	Application is accessible	Use header, footer, and menu links	Navigation works correctly and opens intended pages	UI
TC-25	API	Verify login and search APIs	API endpoints are available	Send valid requests in Postman to login and search endpoints	Correct status codes and expected response fields are returned	API

Bug Report

Bug ID	Title	Severity	Priority	Steps to Reproduce	Expected Result	Actual Result
BUG-01	Login fails with valid credentials	High	High	Open login page, enter valid email and password, click Login	User should be logged in successfully	Error message is displayed and user remains on login page
Bug ID	**Title**	**Severity**	**Priority**	**Steps to Reproduce**	**Expected Result**	**Actual Result**
BUG-02	Search crashes when input is empty	Medium	Medium	Open search page, leave search field blank, click Search	Validation message should appear or page should handle empty input safely	Application crashes or becomes unresponsive
BUG-03	Payment succeeds but booking confirmation is not generated	Critical	High	Select property, complete booking details, make successful payment	Booking confirmation page and confirmation record should be created	Payment succeeds but booking confirmation is missing
BUG-04	Profile update is not saved	Medium	Medium	Open profile page, modify user details, click Save	Updated profile details should be stored successfully	Old profile data remains unchanged after refresh

BUG-05	UI breaks on mobile screen	Low	Medium	Open application on small screen viewport, navigate to search and property page	UI should remain aligned and responsive	Text and buttons overlap or move out of screen
BUG-06	Home page loads slowly above acceptable limit	Medium	Medium	Open home page on normal connection and measure load time	Home page should load within 3 seconds	Page takes more than 5 seconds to load
BUG-07	Logout button does not end user session	High	High	Login successfully, click Logout	User session should end and redirect should happen	Logout click has no effect or session remains active

Automation Testing

Automated Scenarios

- Login functionality
- Booking flow

Selenium Automation Script (Python)

```
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.common.keys import Keys
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC

base_url = "https://example-airbnb-clone.com" driver = webdriver.Chrome() driver.maximize_window() wait = WebDriverWait(driver, 10)

try:
    driver.get(f"{base_url}/login") wait.until(EC.visibility_of_element_located((By.ID, "email"))).send_keys("test@mail.
    driver.find_element(By.ID, "password").send_keys("123456") driver.find_element(By.ID, "login").click() wait.until(EC.url_contains("dashboard")) assert "dashboard" in driver.current_url.lower()

    driver.get(f"{base_url}/search") wait.until(EC.visibility_of_element_located((By.ID, "location"))).send_keys("Goa") driver.find_element(By.ID, "location").send_keys(Keys.ENTER) wait.until(EC.element_to_be_clickable((By.CSS_SELECTOR, ".property-card"))).click() wait.until(EC.element_to_be_clickable((By.ID, "reserve"))).click()
    wait.until(EC.visibility_of_element_located((By.CLASS_NAME, "booking-confirmation") assert "confirmed" in driver.page_source.lower()
finally:
    driver.quit()
```

API Testing

API ID	API Name	Method	Sample Endpoint	Validation Performed	Expected Result
API-01	Login API	POST	/api/login	Verified status code, response body, token or success flag, invalid credential handling	Returns 200 OK for valid credentials and appropriate error for invalid credentials
API-02	Search API	GET	/api/search?location=Goa	Verified status code, data array, empty result handling, response format	Returns 200 OK and matching property data for valid search input

Sample Postman Checks

- Status code is 200 for successful requests.
- Response body contains required fields such as message, user, token, listings, or booking details based on endpoint behaviour.
- Invalid input returns controlled error response instead of server crash.

Final Report

Summary of Testing

End-to-end testing was performed for major Airbnb Clone workflows including registration, login, search, booking, payment, profile management, UI behaviour, and API validation. The assignment includes manual testing, API testing using Postman, and automation of two critical user journeys using Selenium.

Key Findings

- Critical issue found in payment flow where payment succeeds but booking confirmation is missing.
- High-severity defects found in login and logout functionality.
- Search flow needs stronger validation for empty input handling.
- Mobile responsiveness and page performance require improvement before production release.

Suggestions

- Fix all critical and high-severity defects before release.
- Add stronger front-end and back-end validation for login, search, and booking flows.
- Improve responsive layout behavior on smaller screens.
- Optimize page assets and API calls to improve load time.
- Expand regression coverage after every major bug fix.